

Recitation #5

10-606

September 2025

Convexity and Gradient Descent

1. Determine whether the function $f(x, y) = 2x^2 + 3y^2 - xy$ is convex by calculating the Hessian matrix.
2. Show that the L2 regularization term $f(\beta) = \|\beta\|_2^2$ is convex. Use the second derivative test for this function.
3. For the linear regression model $y = X\beta$ with an L2 regularization term $\|\beta\|_2^2$, the objective function becomes $L(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|_2^2$, where $\lambda > 0$ is a regularization parameter. Compute the gradient of $L(\beta)$ with respect to β .
4. Consider the linear regression model $y = X\beta$, where $X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}$ and $y = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$. Starting with $\beta_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, perform one iteration of gradient descent with learning rate $\alpha = 0.01$.

Probability

1. You roll a fair six-sided die.
 - (a) What is the sample space?
 - (b) Define the event “rolling an even number.” What is its probability?
 - (c) Define the random variable $X = (\text{outcome mod } 2)$. Give the distribution of X .
2. A fair coin is flipped 3 times.
 - (a) What is the probability of exactly two heads?
 - (b) What is the probability of at least one head?
 - (c) If the coin were biased with $P(H) = 0.7$, what changes?

3. Two fair dice are rolled. Let X =value of die 1, Y =value of die 2.
- (a) What is $P(X = 3, Y = 5)$?
 - (b) What is $P(X = 3 \mid X + Y = 8)$?
 - (c) Are X and Y independent?